Object Oriented Programming (CSEG1044)

Unit 04 – Lecture 05: Swing Event Handling + JDBC (Persistence)

April 1, 2026

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed April 1, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed April 1, 2026).

Quick Links

Overview Key Topics Demo Summary

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Agenda

1 Overview

2 Key Topics

3 Demo / Example

4 Summary

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Lecture Snapshot

- **Unit focus:** Generics, Lambdas, Swing, and JDBC

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Lecture Snapshot

- **Unit focus:** Generics, Lambdas, Swing, and JDBC
- **Course thread:** Use Swing Event Handling + JDBC (Persistence) to move Campus Resource Manager toward a real desktop application with UI, callbacks, and persistence.

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Recall Before You Start

- **Swing foundations**: You already know the component structure; now user actions and persistence are attached to that UI. Main reference: `Unit 04 / Lecture 04 slides/notes`.

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed April 1, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed April 1, 2026).

Recall Before You Start

- **Swing foundations**: You already know the component structure; now user actions and persistence are attached to that UI. Main reference: `Unit 04 / Lecture 04 slides/notes`.
- **Failure handling**: JDBC and event code both need disciplined exception handling and user messages. Main reference: `Unit 03 / Lecture 02 slides/notes`.

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Learning Outcomes

- Connect Swing UI controls to actions using listeners.

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Learning Outcomes

- Connect Swing UI controls to actions using listeners.
- Explain JDBC flow and choose between `Statement` and `PreparedStatement`.

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Learning Outcomes

- Connect Swing UI controls to actions using listeners.
- Explain JDBC flow and choose between `Statement` and `PreparedStatement`.
- Implement basic CRUD and read results using `ResultSet`.

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Key Terms to Know

- **Event source**: UI component that triggers an event (button, menu item).
- **Event object**: information about the event (e.g., `ActionEvent`).
- **Listener**: object/function that handles events (e.g., `ActionListener`).
- **EDT**: Event Dispatch Thread (Swing UI thread).
- **JDBC**: Java Database Connectivity API.
- **Connection**: active DB connection.
- **PreparedStatement**: parameterized SQL statement (safer and common).
- **ResultSet**: table-like cursor over query results.
- **CRUD**: Create, Read, Update, Delete.
- **DAO**: Data Access Object (separate DB code from UI).

Study Flow for This Lecture

1. Refresh the prerequisite bridge before reading new ideas in isolation.

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1. Refresh the prerequisite bridge before reading new ideas in isolation.
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3. Trace the worked example and connect each line back to one concept frame.

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3. Trace the worked example and connect each line back to one concept frame.
4. Attempt the mini exercises before moving to the recap and exit check.

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Unit 04: Generics, Lambdas, GUI Swing & Database Connectivity

- Lecture focus: Swing Event Handling + JDBC (Persistence)

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Today: Roadmap

- Swing events: listeners, event classes, common controls

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Today: Roadmap

- Swing events: listeners, event classes, common controls
- JDBC core flow: connection, ‘Statement‘, ‘PreparedStatement‘, ‘CallableStatement‘

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Today: Roadmap

- Swing events: listeners, event classes, common controls
- JDBC core flow: connection, ‘Statement‘, ‘PreparedStatement‘, ‘CallableStatement‘
- ‘ResultSet‘ traversal, basic CRUD, and persistent data design

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Swing events (model)

- Source: component (e.g., `JButton`)

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Swing events (model)

- Source: component (e.g., `JButton`)
- Event: object with details (e.g., `ActionEvent`)

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Swing events (model)

- Source: component (e.g., `JButton`)
- Event: object with details (e.g., `ActionEvent`)
- Listener: handler (e.g., `ActionListener`)

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Registering a listener

- `saveBtn.addActionListener(...);`

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Registering a listener

- `saveBtn.addActionListener(...);`
- In Java 8+, a lambda is commonly used for `ActionListener`.

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Registering a listener

- `saveBtn.addActionListener(...);`
- In Java 8+, a lambda is commonly used for `ActionListener`.
- Keep handler short: validate input, call DAO, show message.

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JDBC flow (memorize)

1. Get `Connection`

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JDBC flow (memorize)

1. Get `Connection`
2. Create `PreparedStatement`

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JDBC flow (memorize)

1. Get `Connection`
2. Create `PreparedStatement`
3. Execute SQL (`executeUpdate`/`executeQuery`)

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JDBC flow (memorize)

1. Get `Connection`
2. Create `PreparedStatement`
3. Execute SQL (`executeUpdate`/`executeQuery`)
4. Read `ResultSet` for SELECT

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JDBC flow (memorize)

1. Get `Connection`
2. Create `PreparedStatement`
3. Execute SQL (`executeUpdate`/`executeQuery`)
4. Read `ResultSet` for SELECT
5. Close resources (use `try-with-resources`)

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Statement vs PreparedStatement

- `Statement`: builds SQL strings; risky with user input.

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- `Statement`: builds SQL strings; risky with user input.
- `PreparedStatement`: parameterized (?); safer and common.

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Statement vs PreparedStatement

- `Statement`: builds SQL strings; risky with user input.
- `PreparedStatement`: parameterized (?); safer and common.
- `CallableStatement`: stored procedures (DB-specific).

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ResultSet (reading rows)

- Use cursor loop: `while (rs.next()) {...}`

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ResultSet (reading rows)

- Use cursor loop: `while (rs.next()) {...}`
- Get columns: `rs.getString("name")`, `rs.getInt("capacity")`

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ResultSet (reading rows)

- Use cursor loop: `while (rs.next()) {...}`
- Get columns: `rs.getString("name")`, `rs.getInt("capacity")`
- Map to objects: `Resource` model class

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Checkpoint and Reflection

Reflection prompt

Where would *Swing Event Handling + JDBC (Persistence)* matter most in Campus Resource Manager, and which design choice would you justify using this lecture's ideas?

Quick self-check

Which part needs one more example before you move on: *Swing event model (listeners)*, *JDBC core flow (pattern you should memorize)*, the worked example, or the recap?

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Example: button click handler (save)

```
saveBtn.addActionListener(e -> {
    String name = nameField.getText().trim();
    if (name.isEmpty()) {
        JOptionPane.showMessageDialog(f, "Name required");
        return;
    }
    saveInBackground(name, type, capacity);
});
```

- Listener does quick validation only.

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed April 1, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed April 1, 2026).

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```

- Listener does quick validation only.
- A helper/background task performs the DB insert and shows success/failure.

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Example: JDBC insert (PreparedStatement)

```
String sql = "INSERT INTO resources(name, type, capacity)
    VALUES(?,?,?)";
try (Connection con = DriverManager.getConnection(url,
    user, pass);
     PreparedStatement ps = con.prepareStatement(sql)) {
  ps.setString(1, r.getName());
  ps.setString(2, r.getType());
  ps.setInt(3, r.getCapacity());
  ps.executeUpdate();
}
```

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Mini Exercises (2–3 mins)

1. Which is safer for user input: `Statement` or `PreparedStatement`?

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1. Which is safer for user input: `Statement` or `PreparedStatement`?
2. Which method for SELECT: `executeQuery` or `executeUpdate`?

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Mini Exercises (2–3 mins)

1. Which is safer for user input: `Statement` or `PreparedStatement`?
2. Which method for SELECT: `executeQuery` or `executeUpdate`?
3. Why should UI not freeze during DB operations?

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Watch-outs

- Building SQL strings directly from user input.

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Watch-outs

- Building SQL strings directly from user input.
- Running slow database work on the Event Dispatch Thread.

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Watch-outs

- Building SQL strings directly from user input.
- Running slow database work on the Event Dispatch Thread.
- Mixing UI controls, validation, and DAO logic in one method.

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Key Takeaways

- Event handling connects UI actions to code using listeners.

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Key Takeaways

- Event handling connects UI actions to code using listeners.
- JDBC provides the standard pattern for DB access in Java.

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Key Takeaways

- Event handling connects UI actions to code using listeners.
- JDBC provides the standard pattern for DB access in Java.
- Prefer `PreparedStatement` and use `try-with-resources`.

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed April 1, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed April 1, 2026).

Checkpoint Questions

1. What is a listener and how do you register it?

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Checkpoint Questions

1. What is a listener and how do you register it?
2. Why is `PreparedStatement` preferred?

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Checkpoint Questions

1. What is a listener and how do you register it?
2. Why is `PreparedStatement` preferred?
3. What is `ResultSet` used for?

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Next Study Steps

- Revisit the recall references if any older idea still feels weak.

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- Rework the lecture example without looking at the notes first.

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- Rework the lecture example without looking at the notes first.
- Attempt the self-study practice and then answer the exit questions in full sentences.
- Apply the idea once inside Campus Resource Manager to make the concept stick.

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Next Unit Preview

Next: Unit 05 – Collections Framework

- We will store objects in lists/sets/maps and iterate/sort them.

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